UNIVERSITY OF MORATUWA

Department of Computer Science and Engineering

PROJECT PROPOSAL

Second-Hand Vehicle Marketplace with Intelligent Search and Automated Inventory Processing

Project ID(PID)	11
Group Number	25
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1. Overview of the Project

This project is about a web based B2B marketplace for second hand vehicles. The system allows vehicle dealers to upload inventories in bulk using CSV or JSON files along with ZIP archives of vehicle images. Each vehicle record includes details like make, model, price, mileage, fuel type , associated images etc.

Uploaded inventories go through an automated Extract, Transform Load (ETL) pipeline. In the Extract stage, vehicle data and images are uploaded and stored [8]. In the Transform stage, background workers validate records, standardize vehicle attributes, link images with listings and create search ready metadata. Finally, in the Load stage, validated records are stored in PostgreSQL [6], while searchable metadata and semantic embeddings are indexed using PostgreSQL extensions [7].

Buyers and guests can browse vehicles with standard filters such as make, model etc. Registered buyers can also save favorite vehicles and search using natural language queries, like "I need an automatic SUV under 20 million LKR." The search module interprets the query, extracts important data , and returns relevant vehicles through structured filtering and semantic search.

Inputs	Outputs
Dealer, buyer and admin registration details	Validated and processed vehicle listings
CSV or JSON vehicle inventory files	Optimized vehicle images
ZIP files containing vehicle images	Traditional and intelligent search results
Vehicle search requests	Dealer inventory dashboard
Dealer inventory updates	Upload status reports and error reports
	Email notifications
	Synthetic datasets for testing and development

2. Objectives of the Project

The objectives of this project are to:

- To design a scalable, B2B architecture for bulk vehicle data ingestion and processing.
- To implement an asynchronous Automated ETL (Extract-Transform-Load) Pipeline that validates, transforms , standardizes and enriches vehicle listings with consistent metadata.

- To provide traditional filter based and intelligent natural language vehicle search.
- To develop a dealer dashboard for managing their inventories.
- To deploy the system in local Docker environments and AWS cloud infrastructure.
- To establish a CI/CD process pipeline with automated testing and deployment.

3. Need of the project

Many car dealers still upload vehicle details manually, which takes a lot of time and can cause errors. Vehicle information may be incomplete or inconsistent. Also most locally existing websites only allow buyers to search vehicles using filters. They cannot easily search using normal English sentences.This project targets to solve these problems by providing bulk uploads, automatic data processing, intelligent search and cloud based deployment. It helps dealers manage their inventory more easily and helps buyers find suitable vehicles efficiently. Although existing international products solve some of these issues they are expensive for small scale car dealers.

4. Scope of the Project

The proposed system will include four main user roles.

Dealer - Register/login, Upload vehicle inventories in bulk (CSV/JSON) along with image assets, Track processing pipeline status, Manage active listings via an inventory dashboard, Receive automated email summaries upon upload completion.

Buyer - Register/log in, Browse the vehicle catalogue, Search using Natural Language or filters, View details of vehicles, Save favourite information

Guest - Browse available vehicles, Search vehicles using filters or natural language queries, View details of vehicles, Register or log in to access additional features

Administrator - Manage users (dealers and buyers), Monitor upload jobs and processing pipelines, View system logs and reports, Manage the platform

The Automated ETL Pipeline - Data validation, Data normalization, Image to vehicle mapping, Image resizing, compression and format conversion (if needed), Metadata generation which are used for search, Storage of validated listings, Reporting invalid records, Asynchronous job processing and status tracking

5. Deliverables

- A web based second hand vehicle marketplace
- A Dealer dashboard
- An Admin dashboard
- A buyer and guest vehicle search interface
- Bulk CSV/JSON inventory uploading functionality
- An Automated ETL Pipeline
- Vehicle image upload and optimization module
- Filter based and natural language based search
- Upload status reports and email notifications

- Docker based local development environment
- AWS cloud deployment
- CI/CD pipeline configuration
- Synthetic vehicle data generation tool for testing
- Technical documentation and user documentation

6. Overview of Existing Systems and Technology

There are several online vehicle marketplaces such as AutoTrader [1], Cars.com [2], ikman.lk [3], and Riyasewana [4] that allow users to search and view vehicle listings .However, the international platforms among these are designed for large businesses and may not provide affordable solutions for smaller dealerships. Out of these, the local platforms may focus mainly on traditional search methods and have the manual entry method.Our project will use modern cloud technologies and automation to improve inventory management and provide a better search experience.

Technologies to be Used

React for frontend development [9], **NestJS** for backend services [5], **PostgreSQL** hosted on **AWS RDS** for storage [6], [8], **pgvector**, a PostgreSQL extension, for semantic search [7] and a **Hugging Face sentence embedding model** to generate vectors [13]. **AWS S3** is used for image storage, **AWS SQS** for ETL processing, **AWS SES** for email notifications and **AWS ECS** for containerized deployment [8]. **Docker** is used for containerization [10], **Terraform** for Infrastructure as Code [11] and **GitHub Actions** for CI/CD [12].

7. References

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